Math 1030
Magnets and Pain Relief Statistical Project

Is it possible that magnetic fields can reduce pain? A fascinating study by Dr. Carlos Vallbona suggests it is.

Background

Magnetic fields have been shown to have an effect on living tissue as early as the 1930's. Plants have been shown to have an improved growth rate when raised in a magnetic field (Mericle et al., 1964). More recently, doctors and physical therapists have used either static or fluctuating magnetic fields to aid in pain management, most commonly for broken bones. In the case study presented here, Carlos Vallbona and his colleagues sought to answer the question "Can the chronic pain experienced by postpolio patients be relieved by magnetic fields applied directly over an identified pain trigger point?"

Mericle, R. P. et al. "Plant Growth Responses" in: *Biological effects of magnetic fields.* New York: Plenium Press 1964. p183-195.

Vallbona, Carlos et. al., "Response of pain to static Magnetic fields in postpolio patients, a double blind Pilot study" *Archives of Physical Medicine and Rehabilitation*. Vol 78, American congress of rehabilitation medicine, p 1200-1203.

Experimental Design

Summary

Patients experiencing post-polio pain syndrome were recruited. Half of the patients were treated with an active magnetic device and half were treated with an inactive device. All patients rated their pain before and after application of the device

Details

The experimenters recruited 50 patients who not only had post-polio syndrome but also reported muscular or arthritic pain. These patients had significant pain for at least 4 weeks and had not taken any painkillers or anti-inflammatories for at least 3 hours before the study. The subjects all had a trigger point or painful region and had a body weight of less than 140% of the predicted weight for their age and height, and had a trigger point or circumscribed painful area.

The magnets and placebos (described below) were supplied in equal numbers from Bioflex. Each magnet or placebo was placed in number coded envelopes and delivered according to its shape. The code for placebos and magnets was not broken until the end of the study.

One site of reported pain was evaluated and a trigger point for this pain was found by palpitation. The patient was asked to subjectively grade pain at the trigger point under palpitation on a scale from 1 to 10 (1 is the least pain, increasing to 10).

Following the initial pain assessment, an envelope containing a device was randomly selected from the box containing both active and inactive devices. This device was applied to the pain area for 45 minutes and then removed. The patient then evaluated his or her pain again at the region or trigger point. This second pain rating is the score analyzed here.

Materials

The magnets used in the study were Bioflex magnets. The magnetic field consists of concentrically arranged circles of alternating magnetic polarity. Seventy active devices and seventy identical inactive devices were used.

Relevant Data found in the Study
(For your reference. This will NOT need to be turned in)

Active Magnets

Pre	Post	Change
10	0	10
10	4	6
8	7	1
10	0	10
10	4	6
10	2	8
10	5	5
10	5	5
9	3	6
10	2	8
9	2	7
10	2	8
10	3	7
10	5	5
10	6	4
8	4	4
10	3	7
10	0	10
8	2	6
10	0	10
10	4	6
9	4	5
10	5	5
10	9	1
10	10	0
10	10	0
10	10	0
10	10	0
8	7	1

Placebos

Pre	Post	Change
8	4	4
10	7	3
10	5	5
10	8	2
9	8	1
10	6	4
9	8	1
10	10	0
10	10	0
7	6	1
10	10	0
8	8	0
10	10	0
10	10	0
10	10	0
10	10	0
9	9	0
10	9	1
10	10	0
10	10	0
10	9	1

Magnets & Pain Relief – a Math 1030 Project

Name of Student Submitting the official Assignment: Anna Warr

Names of other Group Members (Best to work in groups of two to four):

Joseph Christiansen

Kailani Uhila

Directions:

You will earn 5 extra points (up to 20 pts) for each additional class member you work with and turn in only one assignment for the group. Note that you are limited to 4 students in a group. You are responsible for creating these groups and checking each other's work. You may create a group by messaging your peers in Canvas or asking on the question board. You can meet in person or online through google, Canvas groups or any other method. Although everyone in the group should submit the assignment, you only get extra points if you turn in the SAME assignment. I will only grade the first submission and then transfer the scores to the rest of the group members.

You will be analyzing three things in this project:

1. The Pretest Information for both the active and placebo group
2. The Posttest Information for both the active and placebo group
3. The Change in Pain for both the active and placebo group

On each of the following pages, the first page should be your synopsis containing the relevant statistics and visualizations of the data derived.

On the back (2^{nd}) page you should show me all of your work. If you are unable to keep the work contained in the box you may attach a separate sheet of paper, but the work needs to be clearly labeled and organized. You may use a calculator, but should still show me what was entered into the calculator.

Lastly, you need to summarize your results. Type a one paragraph essay (minimum of 100 words) and attach to this document reflecting how:

1. These statistics either support, refute, or are irrelevant to the claim that magnets can help relieve some types of pain
2. Discuss the limits for which the data can be used and
3. Why you think this is, or is not, a good study.

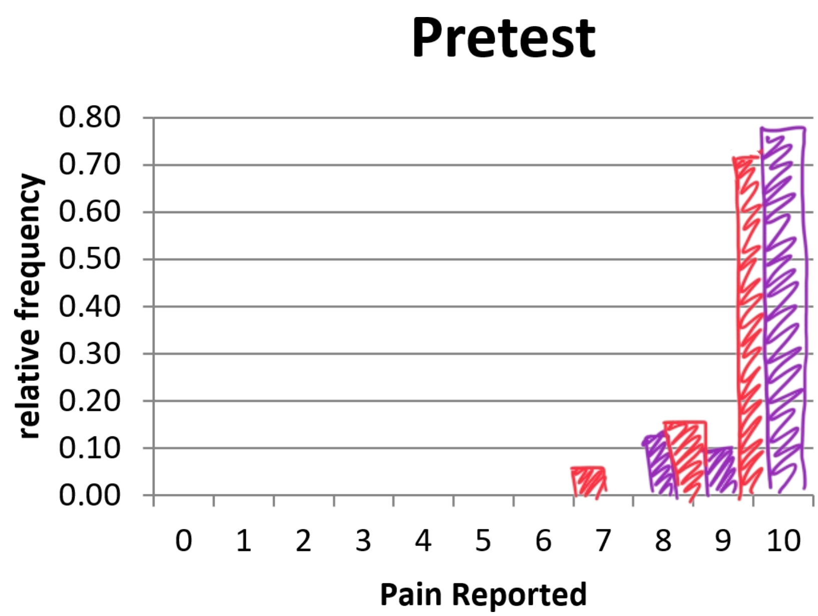
You will submit this cover sheet, along with all remaining pages. (You do not need to submit the former first two pages)

Describing the Data Found in the Pretest Results

Compare the mean, range, and distribution (graphing both a double histogram using **relative frequency** and box plots) of the **pretest** for the active magnet group and the placebo group.

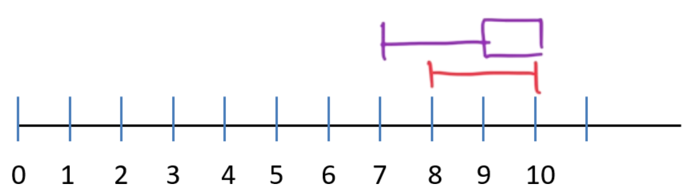
Active Magnets: mean = 9.62
range = 2

Placebo: mean = 9.52
range = 3



Active:

Placebo:



Do the groups appear similar or different on the Pretest? State which and explain ***why*** you think the groups appear to be similar or different using one or two complete sentences.

The groups appear similar. It makes sense they have similar pain levels since this is before any treatments.

Detail Regarding the Pretest Results

Work calculating ***Mean and Range*** for the Active Group:

Mean:

$10+10+8+10+10+10+10+10+9+10+10+10+10+8+10+10+8+10+10+9+10+10+10+10+10+10+8$

$= 279$

$279 \div 29$

$= 9.62$

Range: $10-8=2$

Work calculating ***Mean and Range*** for the Placebo Group:

Mean:

$8+10+10+10+9+10+9+10+10+7+10+8+10+10+10+10+9+10+10+10+10$

$= 200$

$200 \div 21$

$= 9.52$

Range: $10-7=3$

Work for calculating the ***Relative Frequencies*** for the Active Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	4	14%
9	3	10%
10	22	76%

Work for calculating the ***Relative Frequencies*** for the Placebo Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	1	4.8%
8	2	9.5%
9	3	14.3%
10	15	71.4%

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 8

Q1: $0.25(29) = 7.25$
is in the 8th spot

Median: 10

Q3: $0.75(29) = 21.75$
is in the 22nd spot

Max: 10

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 7

Q1: $0.25(21) = 5.25$
$= 9$

Median: 10

Q3: $0.75(21) = 15.75$
$= 10$

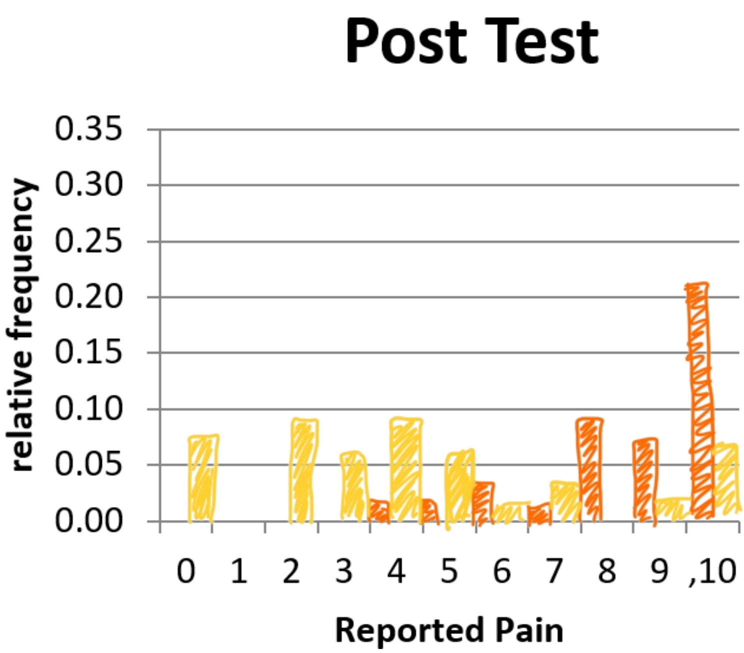
Max: 10

Describing the Data Found in the Post-Test Results

Compare the mean, range, and distribution (using both histograms and box plots) of the **post-test** for the active magnet group and the placebo group.

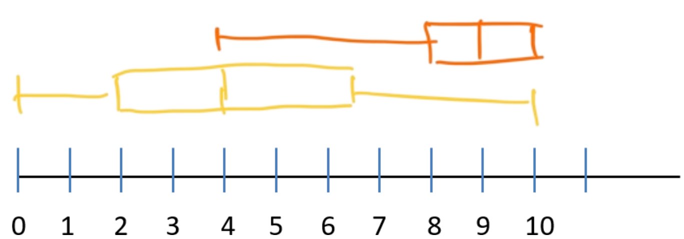
Active Magnets: mean = 4.41
range = 10

Placebo: mean = 8.43
range = 6



Active:

Placebo:



Do the groups appear similar or different on the Post Test? State which and explain ***why*** you think the groups appear to be similar or different using one or two complete sentences.

The groups are pretty spread out. The group with active magnets had better pain management than the placebo group.

Detail Regarding the Post-Test Results

Work calculating ***Mean and Range*** for the Active Group:

Mean:
$4+7+4+2+5+5+3+2+2+2+2+3+5+6+4+3+2$
$+4+4+5+9+10+10+10+10+7$
$= 130$
$128 \div 29$
$= 4.41$

Range: $10-0=10$

Work calculating ***Mean and Range*** for the Placebo Group:

Mean:
$4+7+5+8+8+6+8+10+10+6+10+8+10+10$
$+10+10+9+9+10+10+9$
$= 177$
$177 \div 21$
$= 8.428$

Range: $10-4=6$

Work for calculating the ***Relative Frequencies*** for the Active Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0	4	13.8%
1	0	0
2	5	17.2%
3	3	10.34%
4	5	17.2%
5	3	10.34%
6	1	3.4%
7	2	6.9%
8	0	0
9	1	3.4%
10	4	13.8%

Work for calculating the ***Relative Frequencies*** for the Placebo Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0	0	0
1	0	0
2	0	0
3	0	0
4	1	4.8%
5	1	4.8%
6	2	9.5%
7	1	4.8%
8	4	19%
9	3	14.3%
10	9	42.9%

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 0

Q1: $0.25(29) = 7.25$
Q1 = 2

Median: 4

Q3: $0.75(29) = 21.75$

Max: 10

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 4

Q1: $0.25(21) = 5.25$
Q1 = 8

Median: 9

Q3: $0.75(21) = 15.75$

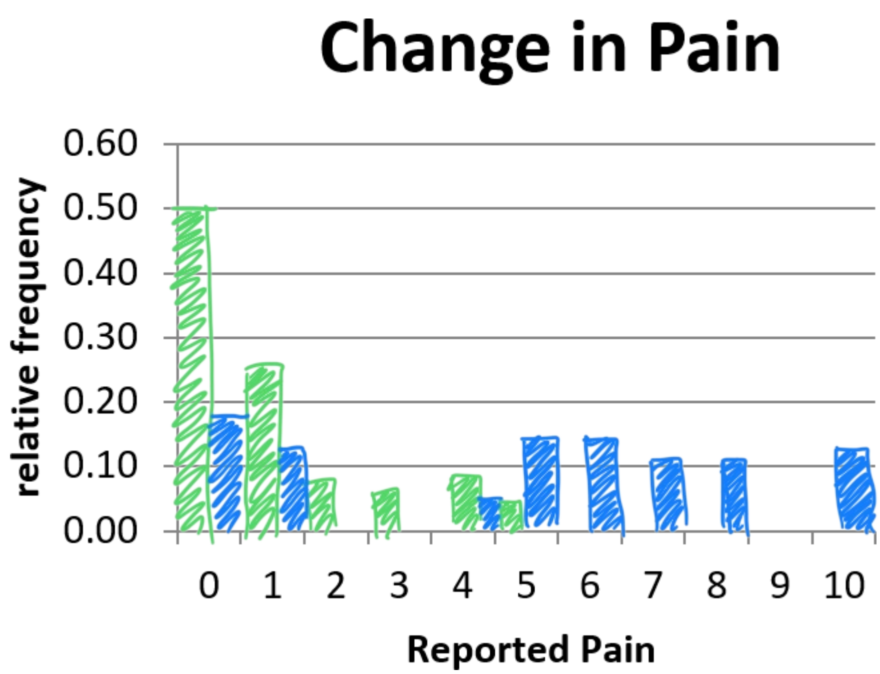
Max: 10

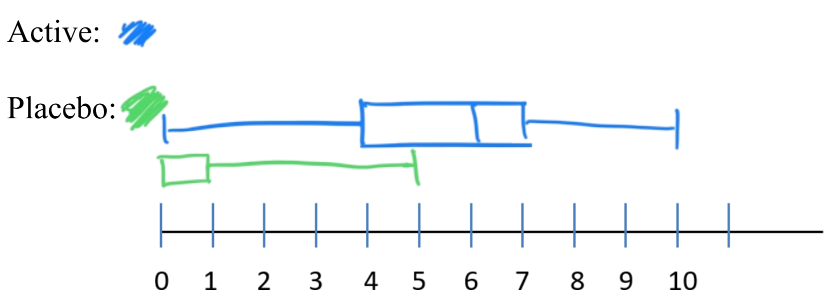
Describing the Data for Change in Pain

Calculate the "**pain relief**," i.e. the **change between the initial pain and that reported after the treatment.** Explain how you calculated the difference and what the numbers mean. Compare the mean, range, and distribution (using both histograms and box plots) of the **change** for the active magnet group and the placebo group.

Active Magnets: mean = 5.21
range = 10

Placebo: mean = 1.095
range = 5





Do the groups appear similar or different in the Change in Pain? State which and explain ***why*** you think the groups appear to be similar or different using one or two complete sentences.

The active magnet group had the greatest change in pain. You can see that by the active group averaging around a 6 versus 0 for placebo.

Detail Regarding the Change in Pain

Work calculating ***Mean and Range*** for the Active Group:

Mean:

10 + 6 + 1 + 10 + 6 + 8 + 5 + 5 + 6 + 8 + 7 + 8 + 7 + 5 + 4 + 4 + 7 + 10 + 6 + 10 + 6 + 5 + 5 + 1 + 1

= 151

151 ÷ 29

= 5.206

Range: 10 - 0 = 10

Work calculating ***Mean and Range*** for the Placebo Group:

Mean:

4 + 3 + 5 + 2 + 1 + 4 + 1 + 1 + 1 + 1

= 23

23 ÷ 21

= 1.0952

Range: 5 - 0 = 5

Work for calculating the ***Relative Frequencies*** for the Active Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0	4	13.8%
1	3	10.3%
2	0	0
3	0	0
4	2	6.9 %
5	5	17.2%
6	5	17.2%
7	3	10.3%
8	3	10.3%
9	0	0
10	4	13.8%

Work for calculating the ***Relative Frequencies*** for the Placebo Group shown on the histogram:

Pain	Frequency	Rel. Freq.
0		52.4%
1		23.8%
2		4.8%
3		4.8%
4		9.5%
5		4.8%
6		0
7		0
8		0
9		0
10		0

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 0

Q1: 0.25(29) = 7.25

Q1 = 4

Median: 6

Q3: 0.75(29) = 21.75

Q3 = 7

Max: 10

State your 5 number summary for the Active Group and show work for Q1 and Q3:

Min: 0

Q1: 0.25(21) = 5.25

Q1 = 0

Median: 0

Q3: 0.75(21) = 15.75

Q3 = 1

Max: 5

This study is evident that magnets are an effective means to manage joint pain. With the difference being greater in the active magnet group than the placebo, it adds to that claim. The study provided good statistics but to be an even more trustworthy study there should have been more quantitative results rather than qualitative, as in the participants ranking their pain verbally. With the study relying vastly on qualitative results it limits the study but it is good regardless.